

Your grade: 92%

Your latest: 92% • Your highest: 92% • To pass you need at least 80%. We keep your highest score.

Next item →

coach

Almost there! Let's review your answers and then try again.

Your work showed skill in understanding pretrained models, LSTM cells, and tensor manipulation, but there is room for growth in implementing LSTM models. Before retrying this assignment, we recommend focusing on this key area:

- LSTM (Coding):** In Question 5, your response missed some key steps involved in implementing an LSTM model to predict the progression of a noisy trigonometric function over time. Try reviewing [this item](#) to refresh your memory.

Good job.

Retry assignment



1. What is the primary purpose of using a pretrained model in neural network training?

1 / 1 point

- ☒ To save time and computational resources by leveraging already learned features.
- ☐ To avoid the need for a validation dataset.
- ☐ To ensure 100% accuracy in classification tasks.
- ☐ To prevent overfitting by using a smaller dataset.

✔ Correct

Correct! Pretrained models help save time and computational resources by using already learned features.

2. When replacing the final classification layers of a pretrained model, what steps should be taken?

1 / 1 point

- ☒ Remove the existing classification layers.

✔ Correct

Correct! The existing classification layers should be removed to adapt the model for the new task.

- ☐ Ensure the new layers have a higher number of neurons than the original layers.

- ☒ Initialize the new layers with random weights.

✔ Correct

Correct! The new layers are often initialized with random weights before training.

- ☐ Set the learning rate to zero for the new layers.

- ☒ Add new layers compatible with the number of classes in the new task.

✔ Correct

Correct! Adding new layers that match the number of classes in the new task is necessary for proper classification.

3. What is an advantage of Long Short-Term Memory (LSTM) cells over basic Recurrent Neural Network (RNN) cells?

1 / 1 point

- ☐ LSTM cells eliminate the need for activation functions
- ☒ LSTM cells can better capture long-term dependencies in data
- ☐ LSTM cells are simpler to implement
- ☐ LSTM cells require less computational power

✔ Correct

Correct! LSTM cells are specifically designed to address the vanishing gradient problem, making them more effective at capturing long-term dependencies.

4. Which PyTorch function is used to change the shape of a tensor without changing its data?

1 / 1 point

- ☐ expand
- ☐ reshape
- ☐ squeeze
- ☒ view

✔ Correct

Correct! The `view` function in PyTorch is used to reshape a tensor without altering its data.

5. Which of the following are steps involved in implementing an LSTM model to predict the progression of a noisy trigonometric function over time?

0.6 / 1 point

- ☒ Training with mean square error loss and Adam optimizer

✔ Correct

Correct! Training the model with the appropriate loss function and optimizer is necessary for effective learning.

- ☒ Model class definition

✔ Correct

Correct! Defining the model class is essential to structure the LSTM network.

- ☒ Using ReLU activation function

✘ This should not be selected

Incorrect. While ReLU is a popular activation function, LSTM models typically use tanh and sigmoid activation functions.

- ☒ Forward function implementation

✔ Correct

Correct! Implementing the forward function is key to defining how data flows through the network.

- ☐ Dataset setup